

Magnetic resonance imaging features of esthesioneuroblastoma in three dogs and one cat

Charlotte Söffler¹; Antje Hartmann²; Daniela Gorgas^{2*}; Eberhard Ludewig^{3*}; Kerstin von Pückler¹; Martin Kramer¹; Martin J. Schmidt¹

¹Small Animal Clinic, Justus Liebig University, Giessen, Germany; ²Vetsuisse Fakultät von Bern, Clinical Radiology, Bern, Switzerland;

³Department of Small Animals, University of Leipzig, Leipzig, Germany

Keywords

Olfactory neuroblastoma, MRI, intracranial tumor, dog, cat

Summary

Objective: Esthesioneuroblastoma is a rare malignant intranasal tumor that originates from the olfactory neuroepithelium of the upper nasal cavity, and can destroy the cribriform plate and expand into the neurocranium. Descriptions of the magnetic resonance features of esthesioneuroblastomas in animals are scarce. The objectives of this study were to report the magnetic resonance imaging features of esthesioneuroblastomas in order to determine distinct imaging characteristics that may help distinguish it from other intracranial tumor types. **Material and methods:** Magnetic resonance images of four patients with confirmed esthesioneuroblastomas were reviewed and compared with previously reported cases. **Results:** The esthesioneuroblastomas appeared as oval-shaped, solitary lesions in the caudal nasal cavity that caused osteolysis of the cribriform plate and extended into the brain in all cases. Signal intensity was variable. Contrast enhancement was mild and varied from homogeneous to heterogeneous. A peripheral cystic component was found in two patients and was reported in only one previous case. Mass effect and white matter edema were marked to severe. Osteolysis of facial bones and extension into the facial soft tissues or retrobulbar space were not present in any of the cases, although this has been reported in the literature. **Conclusion:** A definitive diagnosis of esthesioneuroblastoma based on signal intensity or contrast behavior was not possible. Nevertheless, the presence of a mass in the caudal nasal cavity with extension into the neurocranium seems to be a feature highly suspicious of esthesioneuroblastoma. In contrast to other extra-cranial lesions, the extra-cranial mass was relatively small and destruction of facial bones seems to be rare.

Schlüsselwörter

Olfaktorisches Neuroblastom, MRT, intrakranieller Tumor, Hund, Katze

Zusammenfassung

Gegenstand und Ziel: Das Ästhesioneuroblastom ist ein seltener maligner intranasaler Tumor mit Ursprung im olfaktorischen Epithel der oberen Nasenhöhle, der unter Destruktion der Lamina cribrosa ins Neurocranium einwachsen kann. Beschreibungen magnetresonanztomographischer Merkmale von Ästhesioneuroblastomen bei Tieren gibt es kaum. Ziel der Studie war die Beschreibung magnetresonanztomographischer Merkmale von Ästhesioneuroblastomen, um bestimmte Charakteristika zu bestimmen, die bei der Unterscheidung zu anderen intrakraniellen Tumoren helfen. **Material und Methoden:** Magnetresonanztomographische Bilder von vier Patienten mit bestätigtem Ästhesioneuroblastom wurden evaluiert und mit in der Literatur beschriebenen Fällen verglichen. **Ergebnisse:** Die Ästhesioneuroblastome stellten sich als ovale, solitäre Massen in der kaudalen Nasenhöhle dar, die eine Osteolyse der Lamina cribrosa verursachten und in allen Fällen in das Gehirn einwuchsen. Die Signalintensität war variabel. Die geringgradig ausgeprägte Kontrastmittelanreicherung variierte von homogen zu heterogen. Eine periphere zystenartige Komponente wurde bei zwei Patienten detektiert und in einem Fall in der Literatur beschrieben. Masseneffekt sowie Ödem der weißen Substanz waren mittel- bis hochgradig ausgeprägt. Eine Osteolyse der Gesichtsknochen und eine Extension ins umgebende Weichteilgewebe oder in den Retrobulbärraum wurden nicht festgestellt, sind aber in der Literatur beschrieben. Eine definitive Diagnosestellung des Ästhesioneuroblastoms, basierend auf Signalverhalten oder Kontrastmittelanreicherung, war nicht möglich. **Schlussfolgerung:** Eine Masse in der kaudalen Nasenhöhle mit Infiltration ins Neurocranium ist hochverdächtig für das Vorliegen eines Ästhesioneuroblastoms. Im Gegensatz zu anderen extrakraniellen Läsionen wies diese Neoplasie einen kleinen extrakraniellen Anteil auf und führte selten zu einer Destruktion der Gesichtsknochen.

Correspondence to

Charlotte Söffler
Klinik für Kleintiere
der Justus-Liebig-Universität Gießen
Frankfurter Straße 108
35392 Gießen
Germany
Email: charlotte.soeffler@gmx.de

Magnetresonanztomographische Merkmale des Ästhesioneuroblastoms bei drei Hunden und einer Katze

Tierärztl Prax 2016; 44 (K): 333–340
<http://dx.doi.org/10.15654/TPK-150963>

Received: November 18, 2015

Accepted after revision: March 23, 2016

Epub ahead of print: July 7, 2016

* These authors are listed alphabetically and contributed equally to this study.

Introduction

Esthesioneuroblastoma is a rare malignant intranasal tumor. Several synonyms are mentioned in the human and veterinary literature such as intranasal neuroblastoma, olfactory placode tumor, olfactory neuroblastoma, olfactory neuroepithelioma, olfactory esthesioneuroblastoma, esthesioneurocytoma, esthesioneuroepithelioma and esthesioneuroma (7, 12, 20, 21). The tumor originates from the olfactory neuroepithelium of the upper nasal cavity near the cribriform plate. It expands invasively through the cribriform plate into the frontal sinus and neurocranium with the olfactory bulb and the piriform lobe being most commonly affected (6, 10, 11, 17, 24). Metastases to the cervical lymph nodes, lungs, liver, bones, eyes, parotid gland and central nervous system are reported in human and in veterinary medicine (5, 6, 15, 16, 18, 21).

Clinical signs are unspecific and depend on the disease stage. They include respiratory symptoms such as epistaxis, nasal discharge, upper airway obstruction and acute respiratory distress. Neurologic signs such as lethargy, behavioral changes, decreased proprioception, abducens nerve paralysis and seizures are described. Facial swelling including facial deformation, eye problems (e.g. lacrimation and exophthalmos) and enlarged cervical lymph nodes have been documented in human and veterinary patients. Partial or complete anorexia and weight loss have also been reported (10–12, 15, 17, 20, 23, 24).

Rhinoscopy, radiography, computed tomography and magnetic resonance imaging (MRI) can provide useful information about the tumor and build the basis for characterization and staging (10, 12, 15, 20, 24, 25). Descriptions of the magnetic resonance (MR) appearance of esthesioneuroblastomas in animals are scarce. Concerning signal intensity, variable information exists in veterinary literature (11, 20, 22). Commonly, it is described as an isointense mass in T1-weighted and T2-weighted sequences (11, 20, 22). Edema, hemorrhage and cyst-like structures may be present (11, 13). In humans, cysts along the tumor margin are considered highly suspicious of esthesioneuroblastoma (19). Several differential diagnoses including other neoplasms arising from the nasal cavity as well as primary brain tumors are mentioned in the human and veterinary literature (2, 5, 11, 12). Definitive diagnosis

requires histopathology and immunohistochemistry of tissue samples (13, 20).

The purpose of this case study was to describe the MR imaging characteristics of esthesioneuroblastoma and to compare the findings with the literature in order to determine whether this rare tumor has distinct imaging characteristics that may help to distinguish it from other intracranial tumor types.

Material and methods

Medical records from August 2006 to October 2013 from the Small Animal Clinic of the Justus Liebig University in Giessen were searched for MRI examinations of patients with the confirmed diagnosis of an esthesioneuroblastoma. Two patients were found in which the MRI diagnosis of esthesioneuroblastoma was confirmed histologically. Additionally, one case from the Department of Small Animals of the University Leipzig and one case from the Vetsuisse Faculty Bern, Clinical Radiology, were included.

► Table 1 summarizes the signalment of these patients.

The MR examination included, at minimum, a sagittal T2-weighted sequence, a transverse fluid attenuation inversion recovery (FLAIR) sequence and transverse T1-weighted sequences pre and post intravenous administration of contrast medium. Magnetic resonance images were reviewed by two observers, one board certified radiologist and one radiology resident, and a consensus opinion was formed. The following MRI features were evaluated: lesion number (single or multiple), shape (regular or irregular), contact to the cribriform plate (none, small focal, broad based), location and extracranial extension (nasal cavity, frontal sinus and/or retrobulbar space), size, contour (well-defined or ill-defined), signal intensity (T1-weighted, T2-weighted and FLAIR) in comparison to grey matter, presence and location of cystic lesions (absent, single or multiple, peripheral or central), effect on adjacent brain tissue (mass effect including any type of brain herniation, white matter edema, hydrocephalus), effect on adjacent extra-cranial tissue (bones of the skull, intranasal structures, surrounding soft tissue structures) and contrast enhancement (absent, mild, moderate or strong, heterogeneous, homogeneous or ring enhancement). Contrast enhancement was evaluated in comparison to the signal intensity of the zygomatic salivary gland post intravenous contrast administration. Contrast enhancement was classified as mild, if the uptake was less intense, moderate, if the uptake was similar, and severe, if the uptake was stronger than the enhancement of the zygomatic salivary gland. Magnetic resonance imaging features of the described cases were compared with MRI features previously described for esthesioneuroblastomas.

Table 1 Signalement of the four patients with confirmed esthesioneuroblastoma reported in this study.

Tab. 1 Signalement der vier in dieser Studie beschriebenen Patienten mit bestätigtem Ästhesioneuroblastom

Patient	Species	Breed	Gender	Age (years)	Body weight
1	dog	Afghan Hound	male	6	26.2 kg
2	dog	Magyar Vizsla	female	6	20.0 kg
3	dog	Australian Shepherd	male	9	29.3 kg
4	cat	Domestic Short Hair	female	6	4.0 kg

Results

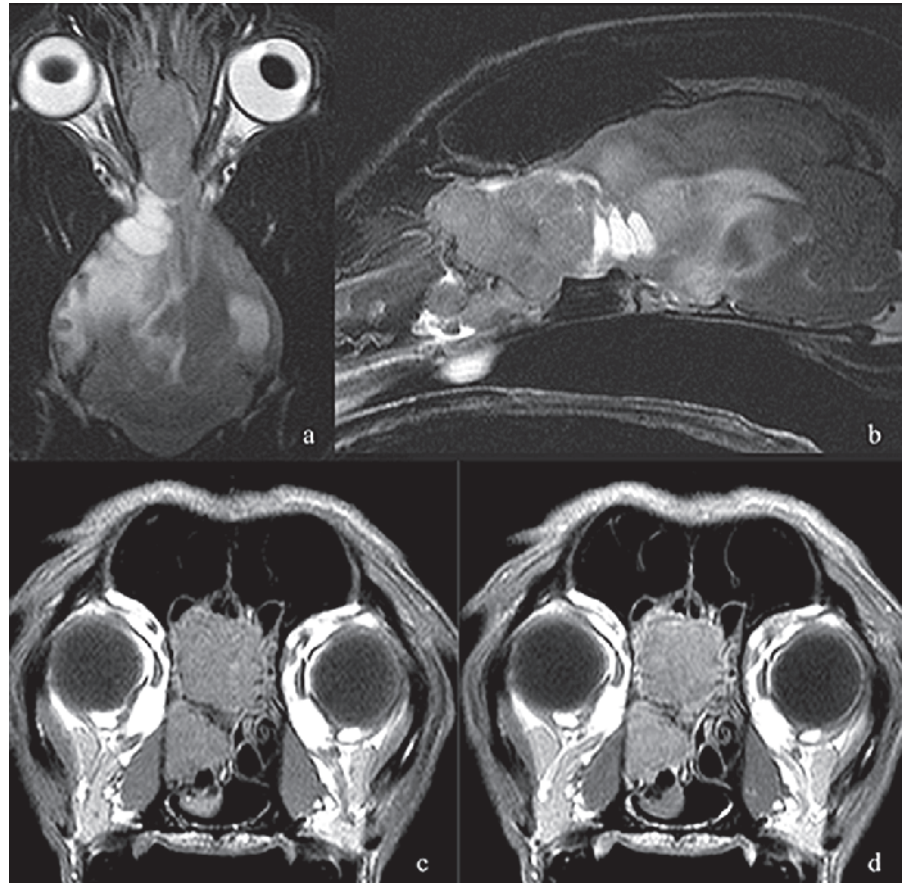
All four animals showed a solitary mass with irregular margins in the caudal nasal cavity extending through the cribriform plate into

Fig. 1

Magnetic resonance images of patient 1, dorsal (a) and sagittal T2-weighted image (b), transverse T1-weighted image pre (c) and post (d) contrast medium application. Esthesioneuroblastoma extending from the right caudal nasal cavity into the olfactory bulb, the piriform lobe and the frontal lobe; severe white matter edema and severe midline shift; cystic structure caudally visible; moderate heterogeneous contrast enhancement of the mass.

Abb. 1

Magnetresonanztomographische Bilder von Patient 1, dorsales (a) und sagittales T2-gewichtetes Bild (b), transversales T1-gewichtetes Bild vor (c) und nach (d) Kontrastmittelgabe. Ästhesioneuroblastom mit Ausdehnung von der rechten Nasenhöhle in den Bulbus olfactorius, den Lobus piriformis und den Lobus frontalis; hochgradiges Ödem der weißen Substanz und hochgradiger Masseneffekt; kaudal zystische Struktur erkennbar; mittelgradige heterogene Kontrastmittelanreicherung der Masse.



the olfactory bulb. In all patients, the mass showed broad based contact to the cribriform plate. The size of the mass was variable. In all patients, the intra-axial component was larger than the extra-axial, intranasal component of the tumor. The approximate ratio between the intra-axial and the extra-axial, intranasal portion of the tumor was 1:1 in patient 1, 90:1 in patient 2, 3:1 in patient 3 and 50:1 in patient 4.

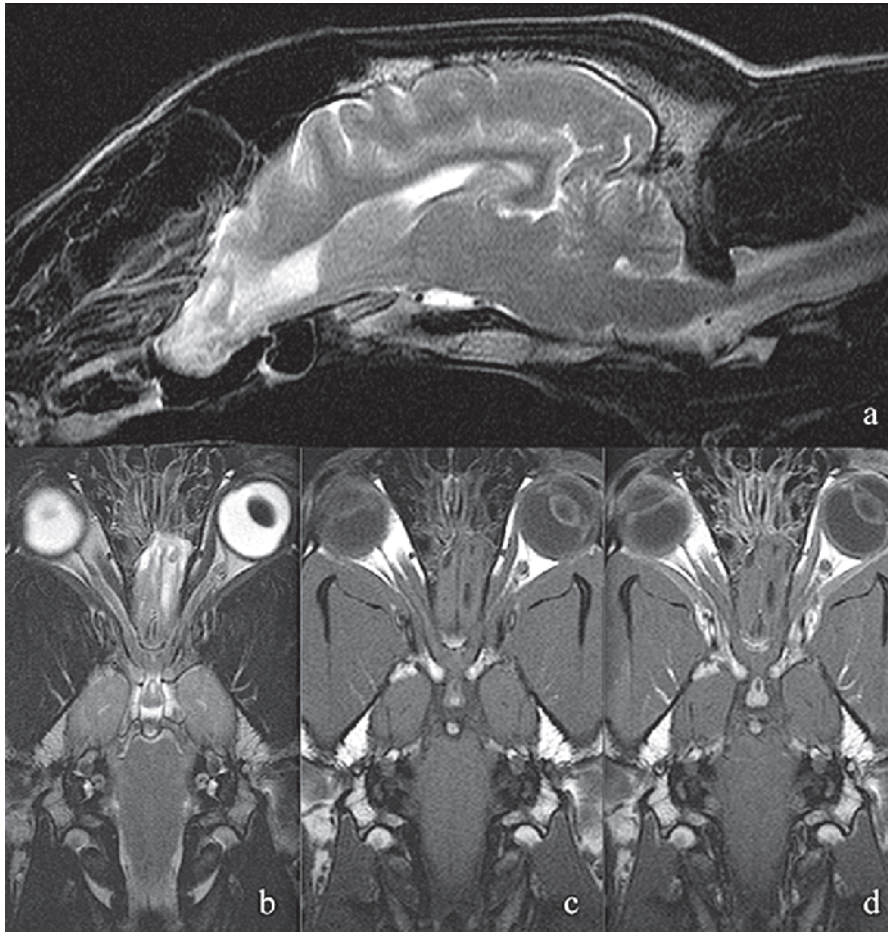
The mass was hypo- to isointense in T1-weighted sequences and hyperintense in T2-weighted and FLAIR sequences, with a mild heterogeneous appearance in all cases. Cystic components were present in two cases. The cystic components were isointense to cerebrospinal fluid in T1- and T2-weighted sequences, but showed no suppression in FLAIR sequences, remaining hyperintense compared to grey matter. In both cases the cysts were located at the caudal border of the mass. White matter edema as well as mass effect on brain tissue varied from mild to severe. Two patients (patient 1 and 4) showed subtentorial herniation of the cerebrum into the caudal fossa. Patient 4 showed additional herniation of the cerebellum through the foramen magnum. Hydrocephalus was present in one dog (patient 1).

Contrast enhancement was variable, ranging from no enhancement to moderate, homogeneous or heterogeneous contrast enhancement (► Figs. 1–4). None of the masses caused osteolysis of the facial bones or extended into the surrounding soft tissues or

retrobulbar space. However, one mass caused a small focal destruction of the ventral lamina of the ethmoid bone and extended slightly into the nasopharyngeal meatus. Imaging features of the four cases are summarized in ► Table 2.

Discussion

The esthesioneuroblastomas in this descriptive case series presented all as solitary, ovoid masses with irregular borders in the caudal nasal cavity. All herein presented cases showed a broad based contact of the mass with the cribriform plate and extension into the brain. This is in accordance with reported cases of esthesioneuroblastoma in veterinary medicine, which describe an extension of the tumor via the cribriform plate into the brain parenchyma in most animals (11, 20, 22). Magnetic resonance imaging features from previously reported cases of esthesioneuroblastomas are summarized in ► Table 3. Osteolysis of facial bones seems to be a rare finding in esthesioneuroblastomas in contrast to other nasal tumors, such as adenocarcinoma, squamous cell carcinoma, or chondrosarcoma (3, 11, 22). Only one report describes extension of the tumor into the retrobulbar space with osteolysis of the maxillary, nasal and frontal bones (20). Another report describes an esthesioneuroblastoma in the nasopharynx,

**Fig. 2**

Magnetic resonance images of patient 2, sagittal (a) and dorsal T2-weighted image (b), dorsal T1-weighted image pre (c) and post (d) contrast medium application. Esthesioneuroblastoma extending from the left caudal nasal cavity into the left olfactory bulb; mild white matter edema, mild dilation of the olfactory recess and mild midline shift; no contrast enhancement.

Abb. 2

Magnetresonanztomographische Bilder von Patient 2, sagittales (a) und dorsales T2-gewichtetes Bild (b), dorsales T1-gewichtetes Bild vor (c) und nach (d) Kontrastmittelgabe. Ästhesioneuroblastom mit Ausdehnung von der linken kaudalen Nasenhöhle in den linken Bulbus olfactorius; geringgradiges Ödem der weißen Substanz, geringgradige Dilatation des Recessus olfactorius und geringgradiger Masseneffekt; keine Kontrastmittelanreicherung.

without contact to the cribriform plate (16). Osteolysis of facial bones and extension into the facial soft tissues were not found in any of the cases of this study. However, one of the herein reported cases showed a small focal osteolytic area of the ventral lamina of the ethmoid bone with mild extension of the mass into the nasopharyngeal meatus.

Signal intensity in our cases as well as in the reported cases is variable, ranging from iso- to hypointense in T1-weighted sequences and from iso- to hyperintense in T2-weighted sequences (11, 20, 22). Thus, differentiation of esthesioneuroblastomas from other nasal or brain tumors is not possible based on signal intensity. Signal intensity is not specific for a certain tumor type (1, 11, 22).

The presence of peripherally located cysts is considered highly suspicious of esthesioneuroblastoma in humans (19). However, in veterinary medicine a cystic component is described in only one of the published cases (11) and only two of the four esthesioneuroblastomas of the present study showed a cystic structure. In addition, up to 30% of neoplastic brain lesions are described to contain cystic structures (4). Hemorrhage can occur (7, 22), but was not visible in any of our cases and seems therefore to be an inconsistent finding, which is probably more related to secondary changes.

White matter edema and mass effect on adjacent brain tissue, with or without herniation of brain structures, were moderate to severe in the published cases (11, 20, 22), consistent with the herein presented cases. There is no relation between the degree of white matter edema to the size of the intracranial portion of the mass, as patient 1 and 3 showed similar sized masses, but a variable degree of white matter edema. Mass effect and brain herniation seem to be related primarily to the degree of white matter edema, and not to the size of the tumor.

Similar to our cases, a strong contrast enhancement, sparing cystic or necrotic parts of the tumor, is described (20, 22). Differential diagnoses include nasal neoplasia (e.g. adenocarcinoma, squamous cell carcinoma, chondrosarcoma) as well as primary brain tumors (e.g. meningioma, oligodendroglioma, glioblastoma, glioma) (2, 5, 8, 11, 12). Due to the commonly large intranasal component, primary nasal tumors are considered the primary differential diagnosis (11, 22). Additionally, nasal tumors are described to invade the intracranial cavity in up to 27.7% of cases (11). Extension of a primary brain tumor into the nasal cavity has not been described yet.

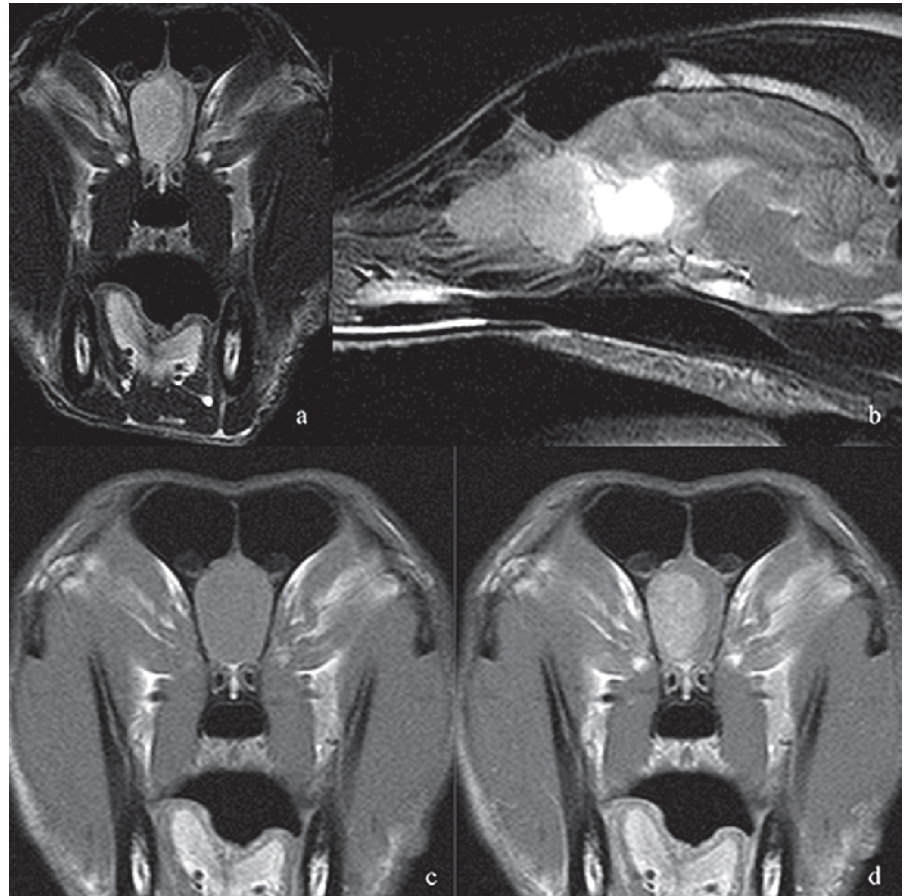
Treatment protocols for primary nasal tumors vary depending on the tumor type. In general, a combination of surgery and irra-

Fig. 3

Magnetic resonance images of patient 3, transverse (a) and sagittal T2-weighted image (b), transverse T1-weighted image pre (c) and post (d) contrast medium application. Esthesioneuroblastoma extending from the right caudal nasal cavity into the olfactory bulb and into the piriform lobe; infiltrative growth with destruction of the nasal conchae; moderate edema and moderate mass effect, cystic structure caudally visible; moderate homogeneous contrast enhancement of the mass and ring enhancement of the cystic lesion.

Abb. 3

Magnetresonanztomographische Bilder von Patient 3, transversales (a) und sagittales T2-gewichtetes Bild (b), transversales T1-gewichtetes Bild vor (c) und nach Kontrastmittelgabe (d). Ästhesioneuroblastom mit Ausdehnung von der rechten kaudalen Nasenhöhle in den Bulbus olfactorius und den Lobus piriformis; infiltratives Wachstum mit Destruktion der Conchae nasales; mittelgradiges Ödem und mittelgradiger Masseneffekt; kaudal zystische Struktur erkennbar; mittelgradige homogene Kontrastmittelanreicherung durch die Masse, periphere Kontrastmittelanreicherung durch die zystische Läsion.

**Fig. 4**

Magnetic resonance images of patient 4, transverse (a) and sagittal T2-weighted image (b), transverse FLAIR image (c), transverse T1-weighted image pre (d) and post (e) contrast medium application. Esthesioneuroblastoma extending from the left caudal nasal cavity into both olfactory bulbs, and into the frontal lobe and temporal lobe, until the level of the thalamus; severe white matter edema and severe mass effect; severe heterogeneous contrast enhancement.

Abb. 4

Magnetresonanztomographische Bilder von Patient 4, transversales (a) und sagittales T2-gewichtetes Bild (b), transversales FLAIR-Bild (c), transversales T1-gewichtetes Bild vor (d) und nach (e) Kontrastmittelgabe. Ästhesioneuroblastom mit Ausdehnung von der linken kaudalen Nasenhöhle beidseits in den Bulbus olfactorius, in den Lobus frontalis und den Lobus temporalis bis auf Höhe des Thalamus; hochgradiges Ödem der weißen Substanz und hochgradiger Masseneffekt; hochgradige heterogene Kontrastmittelanreicherung.

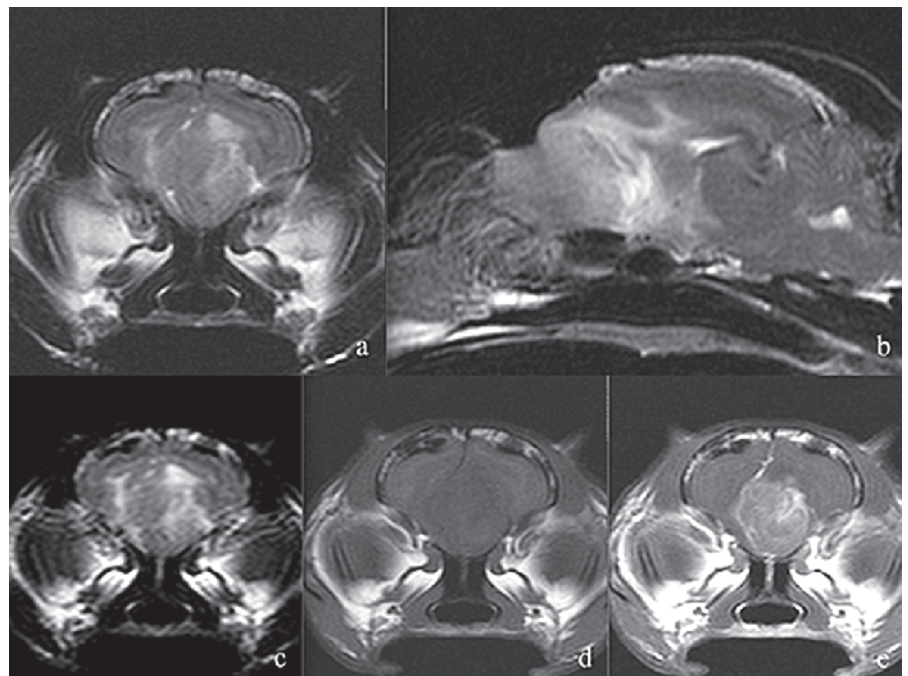


Table 2 Selected magnetic resonance characteristics of the four esthesioneuroblastomas reported in this study.**Tab. 2** Ausgewählte magnetresonanztomographische Merkmale der vier in dieser Studie beschriebenen Ästhesioneuroblastome

Imaging features	Patient			
	1	2	3	4
Lesion number	solitary	solitary	solitary	solitary
Extension into the brain	yes	yes	yes	yes
Osteolysis of facial bones	no	no	no	no
Signal intensity compared to grey matter	T1w: isointense T2w: hyperintense FLAIR: hyperintense, mildly heterogeneous	T1w: isointense T2w: hyperintense FLAIR: hyperintense STIR: hyperintense, homogeneous	T1w: hypointense T2w: hyperintense FLAIR: hyperintense, mildly heterogeneous	T1w: hypo- to isointense T2w: mildly hyperintense FLAIR: mildly hyperintense, mildly heterogeneous
Presence of cyst	yes	no	yes	no
Mass effect on brain tissue	severe	mild	moderate	severe
White matter edema	severe	mild	moderate	severe
Contrast enhancement	moderate, heterogeneous	none	moderate, homogeneous, cyst: ring enhancement	severe, heterogeneous
Additional	small focal destruction of ventral lamina of the ethmoid bone, mild extension into the nasopharyngeal meatus, mild dilation of the right lateral ventricle	mild dilation of the olfactory recess		

FLAIR = fluid attenuation inversion recovery sequence, STIR = short tau inversion recovery sequence, T1w = T1-weighted sequence, T2w = T2-weighted sequence

Table 3 Selected magnetic resonance characteristics of esthesioneuroblastomas previously published in veterinary literature.**Tab. 3** Ausgewählte magnetresonanztomographische Merkmale von Ästhesioneuroblastomen, die in der veterinärmedizinischen Literatur beschrieben sind

Imaging features	Steinmetz et al. (20)				Troxel et al. (22)		Kitagawa et al. (11)
	1	2	3	4	5	6	7
Lesion number	solitary	solitary	solitary	solitary	solitary	solitary	solitary
Extension into the brain	yes	yes	yes	yes	yes	yes	yes
Osteolysis of facial bones	yes (orbit)	yes (orbit)	no	yes (orbit)	no	no	no
Signal intensity	T1w + T2w: isointense	T1w + T2w: isointense	T1w + T2w: isointense	T1w + T2w: isointense	T1w + T2w: hyperintense	T1w: hypointense T2w: isointense	T1w + T2w: isointense
Presence of cyst	no	no	no	no	no	no	yes
Mass effect on brain tissue	yes	yes	no	no	yes	yes	yes
White matter edema	yes	severe	no	no	moderate to severe	moderate to severe	no
Contrast enhancement	strong	strong	strong	strong	strong, heterogeneous	strong, heterogeneous	yes

T1w = T1-weighted sequence, T2w = T2-weighted sequence

diation is recommended in case of sarcomas and carcinomas, whereas chemotherapy is the treatment of choice in case of lymphoma (9, 14, 18, 24). The prognosis for nasal neoplasia is guarded (9). Without treatment, the median survival time for canine nasal carcinomas as well as for sarcomas is approximately 3 months (9, 14). With irradiation, this time is prolonged to approximately 12–23 months (9, 14, 24). The median survival time for cats with intranasal lymphoma, treated with radiation therapy and chemotherapy, is described as approximately 31 months (9). Information about treatment and prognosis of esthesioneuroblastomas is very limited in veterinary medicine. Surgical resection following radiation therapy is described (18, 24). Although clinical signs initially regressed, one dog died at day 448 (24) and the other dog presented regrowth 8 months later and the dog was euthanized (18).

Esthesioneuroblastomas present an aggressive and infiltrative growth (20, 23). A study containing two cats and four dogs with esthesioneuroblastoma demonstrated metastasis in three patients and all patients were euthanized due to a fatal prognosis (20). Outcome of the herein reported cases was poor as well. Two dogs (patient 2 and 3) were euthanized immediately after diagnosis. One dog (patient 1) was treated with surgery followed by irradiation. Euthanasia was performed 8 months later due to recurrence of clinical signs and regrowth of the tumor. The cat (patient 4) went to surgery as well, however 2 weeks after surgery the cat deteriorated. A second MR examination showed regrowth of the mass, which was of similar appearance as on the initial examination and the cat was euthanized. It seems that esthesioneuroblastomas show a poorer outcome, even if treated, compared to primary intranasal tumors after treatment. However, drawing conclusions about prognosis and survival times of esthesioneuroblastomas, based on the small number of cases, is not possible. It therefore seems desirable to improve the differentiation of esthesioneuroblastoma from other neoplasias.

In conclusion, the differentiation between esthesioneuroblastomas and other intracranial neoplasms is challenging. A definitive diagnosis of esthesioneuroblastoma based only on signal intensity or contrast behavior is not possible, as both are not specific for a certain tumor type. Furthermore, cystic structures were not a reliable predictive sign for esthesioneuroblastomas, which differs from human medicine. However, the feature of peripheral cystic components should raise the suspicion of esthesioneuroblastoma, despite the fact that these were not present in every case. An intra-axial tumor without a connection to the nasal cavity should be considered unlikely to be an esthesioneuroblastoma. As esthesioneuroblastomas arise from the olfactory neuroepithelium of the upper nasal cavity and only secondarily invade the brain, an extra-axial mass, outside the calvarium, seems to be always present (6, 10, 11, 20, 24). Primary extra-cranial tumors (e.g. nasal tumors or retrobulbar tumors) cannot be differentiated from esthesioneuroblastomas with reliability. However, those tumors most commonly present as a large extra-cranial mass before invading the brain and often cause osteolysis of the facial bones, which is less likely with esthesioneuroblastomas.

Conclusion and clinical relevance

Differentiation between esthesioneuroblastomas and other intracranial neoplasms is challenging. A definitive diagnosis based on signal intensity or contrast behavior was not possible. Nevertheless, the presence of a mass in the caudal nasal cavity with extension into the neurocranium seems to be a feature highly suspicious of esthesioneuroblastoma.

Conflict of interest

The authors certify that there is no conflict of interest.

References

- Bentley RT. Magnetic resonance imaging diagnosis of brain tumors in dogs. *Vet J* (London, England: 1997) 2015; 205 (2): 204–216.
- Brosinski K, Janik D, Polkinghorne A, Bomhard von W, Schmahl W. Olfactory neuroblastoma in dogs and cats – A histological and immunohistochemical analysis. *J Comp Pathol* 2012; 146 (2–3): 152–159.
- Burk RL. Computed tomographic imaging of nasal disease in 100 dogs. *Vet Radiol Ultrasound* 1992; 33 (3): 177–180.
- Cherubini GB, Mantis P, Martinez TA, Lamb CR, Cappello R. Utility of magnetic resonance imaging for distinguishing neoplastic from non-neoplastic brain lesions in dogs and cats. *Vet Radiol Ultrasound* 2005; 46 (5): 384–387.
- Döpke C, Gröne A, Borstel M, Oppen T, Boëve M, Baumgärtner W. Metastatic esthesioneuroblastoma in a horse. *J Comp Pathol* 2005; 132 (2–3): 218–222.
- Dulguerov P, Calcaterra T. Esthesioneuroblastoma: The UCLA experience 1970–1990. *Laryngoscope* 1992; 102 (8): 843–849.
- Hara K, Shimada A, Morita T, Sawada M, Umemura T. Olfactory neuroepithelioma in a dog: An immunohistochemical and electron microscopic study. *J Vet Med Sci* 2002; 64 (4): 391–393.
- Hecht S, Adams WH. MRI of brain disease in veterinary patients Part 2: Acquired brain disorders. *Vet Clin North Am Small Anim Pract* 2010; 40 (1): 39–63.
- Henry CJ, Higginbotham ML. Cancer management in small animal practice. Maryland Heights, Mo.: Saunders/Elsevier 2010; 203–210.
- Kadish S, Goodman M, Wang CC. Olfactory neuroblastoma – A clinical analysis of 17 cases. *Cancer* 1976; 37 (3): 1571–1576.
- Kitagawa M, Okada M, Yamamura H, Kanayama K, Sakai T. Diagnosis of olfactory neuroblastoma in a dog by magnetic resonance imaging. *Vet Rec* 2006; 159: 288–289.
- Knöbber D, Lobeck H, Schnoy N. Ästhesioneuroblastom: Diagnosesicherung durch Immunhistologie und Elektronenmikroskopie. *Laryngo-Rhino-Otol* 1993; 72 (08): 391–397.
- Kromhout K, Gielen I, Van Caelenberg A, Van Ham L, Van Bree H. Magnetische resonantie bij de detectie van intracraniale letsels bij drie katten. *Vlaams Diergeneeskundig Tijdschrift* 2009; 78 (2): 121–128.
- Kudnig ST, Séguin B, Carlson D. Veterinary surgical oncology. Chichester [England]: Wiley-Blackwell 2012; 87–92, 273–281.
- Morita AM, Ebersold MJ, Olsen KD, Foote RL, Lewis JE, Quast LM. Esthesioneuroblastoma: Prognosis and management. *Neurosurgery* 1993; 32 (5): 706–715.
- Parker VJ, Morrison JA, Yaeger MJ. Olfactory neuroblastoma in a cat. *JFMS* 2010; 12 (11): 867–871.
- Schwartzporsche D, Zwicker C, Weiler H, Stolle A. Tumor of the olfactory mucosa (esthesioneuro-epithelioma) with CNS-invasion in dogs – A case-report with nuclearspintomographic demonstration of the tumor. *Kleintierprax* 1992; 37 (10): 679–686.

18. Siudak K, Klingler M, Schmidt MJ, Herden C. Metastasizing esthesioneuroblastoma in a dog. *Vet Pathol* 2015; 52 (4): 692–695.
19. Som PM, Lidov M, Brandwein M, Catalano P, Biller HF. Sinonasal esthesioneuroblastoma with intracranial extension: marginal tumor cysts as a diagnostic MR finding. *Am J Neuroradiol* 1994; 15 (7): 1259–1262.
20. Steinmetz A, Weber M, Jurina K, Ludewig E, Schoon H, Oechtering G. Das Olfaktoriustumorblastom – Symptomatik, Diagnostik, pathologische Befunde bei zwei Katzen und vier Hunden. *Tierärztl Prax* 2004; 32 (K): 71–81.
21. Thompson LDR. Olfactory neuroblastoma. *Head and Neck Pathol* 2009; 3 (3): 252–259.
22. Troxel MT, Vite CH, Massicotte C, McLear RC, van Winkle TJ, Glass EN et al. Magnetic resonance imaging features of feline intracranial neoplasia: Retrospective analysis of 46 cats. *J Vet Int Med* 2004; 18 (2): 176.
23. Troxel MT, Vite CH, van Winkle TJ, Newton AL, Tiches D, Dayrell-Hart B et al. Feline intracranial neoplasia: Retrospective review of 160 cases (1985–2001). *J Vet Int Med* 2003; 17 (6): 850.
24. Ueno H, Kobayashi Y, Yamada K. Olfactory esthesioneuroblastoma treated with orthovoltage radiotherapy in a dog. *Australian Vet J* 2007; 85 (7): 271–275.
25. Van Meervenne SAE, de Vos JP, Van Ham LML. Therapy of brain tumors in dogs and cats. *Vlaams Diergeneeskundig Tijdschrift* 2007; 76 (3): 165–176.

Anzeige